

Andrea Galenda

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🌐 github.com/andreagalenda | 📍 Pasadena (CA), United States | 📅 01/03/2002

PROFILE

Aerospace-focused Mechanical Engineer specializing in multi-physics simulation, thermodynamics, and disturbance modeling. Proficient in developing comprehensive aero-thermal frameworks in MATLAB and Python to assess dynamic system behavior and operational limits. Distinguished by analytical rigor and a proven ability to extract actionable insights from complex simulation data to support performance analysis and mission evaluation.

EXPERIENCE

NASA Jet Propulsion Laboratory

Dec. 2025 – Present

Visiting Student Researcher - Aerial Robotic Systems (Section 358M)

Pasadena (CA), United States

- Developed a comprehensive aero-thermal flight dynamics model and simulator (MATLAB/Python) to simulate the Venus Aerobot's dynamics, accurately predicting trajectory, altitude, and thermal responses
- Modeled complex atmospheric and environmental perturbations to assess their impact on the Aerobot's dynamic response, extracting actionable insights to evaluate performance limits and operational boundaries
- Formulated a 6-DOF rigid body model and integrated high-fidelity environmental data, incorporating super-rotating wind circulation and cyclostrophic balance effects into a generalized buoyancy model
- Tracked dynamic mass breakdown, leakage flow models, and enthalpy transport, evaluating center of buoyancy offsets and thermodynamic gas expansion to continuously assess aircraft mass configurations and limits

ACADEMIC PROJECTS

Operational Modal Analysis for Health Assessment of Tie Rods

- Reconstructed the Modal Parameters and Frequency Response Function, leveraging the Frequency Domain Decomposition (FDD) algorithm on measurements of the system under natural excitation (NExT)
- Analyzed test data to evaluate dynamic behavior, successfully isolating the variation of modal parameters due to progressive damage on the structure from those due to thermal effects and disturbances

Reduced Order Modeling (ROM) for Complex Physics Simulations

- Processed datasets from finite element simulations using Python to evaluate system deformation and structural boundaries under varying dynamic loads
- Accelerated simulation and data analysis workflows by developing data-driven Reduced Order Models (NN, POD-Galerkin), leveraging Singular Value Decomposition to extract dominant dynamic modes from high-fidelity physics data

EDUCATION

Politecnico di Milano

Sept. 2023 – Present

MSc. in Mechanical Engineering (Track: Mechatronics and Robotics)

Milan, Italy

- **Relevant coursework:** Mechanical Systems Dynamics, Vibrations Analysis, Signal Processing, Advanced Machine Design, Actuating Devices, Electrical Drives, Control Systems
- **Weighted average:** 29.2/30

Polytech Paris-Saclay

Sept. 2024 – Feb. 2025

Erasmus+ Program, Electronics and Robotics Engineering

Paris, France

- **Relevant coursework:** Mechatronics, Computer Vision (C++), AI for Robotics (Python)

Politecnico di Milano

Sept. 2020 – Sept. 2023

BSc. in Mechanical Engineering

Milan, Italy

SKILLS

Simulation and Physics: MATLAB/Simulink, Python, Flight physics modeling, Thermodynamics, System dynamics

Data Analysis and scripting: Python, C++, Signal Processing

Languages: English (C1), French (B2 - Reading-Listening-Speaking), Italian (Native)